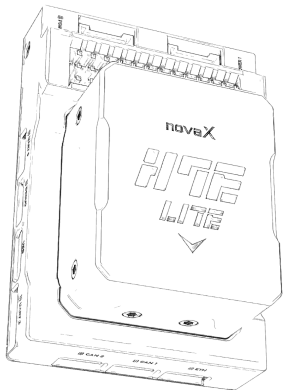


AF-H7E Lite

User Guide



AF-H7E Lite Overview



Compact carrier flight controller that reuses the AF-H7E sensor module (in development). The pin definition may change before release.

AF-H7E Lite Specifications

• Basic Specifications

MCU	STM32H753, ARM Cortex-M7, 480 MHz
RAM / Flash	1 MB / 2 MB
IMU	2× — ICM-42688-P (TDK InvenSense) + BMI088 (Bosch)
IMU Heater	Built-in heater on the IMU board, held at 45 °C (BRD_HEAT_TARG)
Magnetometer	RM3100 (PNI)
Barometer	2× ICP-20100 (TDK InvenSense)
Operating Voltage	4.75 – 5.7 V (Rated 5 V)
USB Input	4.75 – 5.25 V

AF-H7E Lite Specifications

• Basic Specifications

Power Input	POWER 1 + POWER 2 (redundant, I2C power modules)
PWM Output	12 FMU channels (DSHOT except M7–M8) + SBUS out
RC Input	S.Bus, PPM, DSM / Spektrum (with DSM bind)
RSSI Input	Analog
UART	6 serial ports (TELEM 1 · 2, GPS 1 · 2, SERIAL4, SERIAL6) + debug console
Serial Mapping	SERIAL1 = UART7 · SERIAL2 = UART5 · SERIAL3 = USART1 · SERIAL4 = UART8 · SERIAL5 = USART2 · SERIAL6 = UART4 · SERIAL7 = USART3 · SERIAL8 = USART6 (USB = SERIAL0)

AF-H7E Lite Specifications

• Basic Specifications

GPS	GPS 1 + GPS 2 (6-pin, UART + I2C, left / right)
I ² C	2 dedicated ports (I2C A, I2C B) + GPS 1 · 2 + power-module bus on POWER 1 · 2
CAN	2 Port
Ethernet	100 Mbps x 1 Port
Size	42.7 × 63.2 × 29.2 mm (concept case)
Mount Hole	16 × 30 mm / M3 (4 blind holes in the base, 3 mm thread)
Supported F/W	novaX ArduPilot (Copter and Plane releases)

AF-H7E Lite Pinout

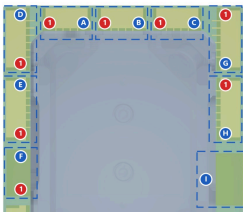
• Pinout (1/4)

AF-H7E Lite Pinout — Top View

Concept layout seen from above. Most connectors sit on the bottom side and plug in from the edges — those maps draw the PCB see-through. Each table reads left → right (side edges: top → bottom) as the map above it.

A = connector, its table below **1** = pin 1, the red 1 in its table + = 5 V - = GND blue = signal

Front half — bottom-side connectors, seen through the PCB (front edge at the top)



A ETHERNET

4-Pin · JST-GH

1	2	3	4
RX-	RX+	TX-	TX+

B CAN 1

4-Pin · JST-GH

1	2	3	4
+	CAN_H	CAN_L	-

C CAN 2

4-Pin · JST-GH

1	2	3	4
+	CAN_H	CAN_L	-

D TELEM 1

SERIAL1 · 6-Pin · JST-GH

6	-
5	RTS
4	CTS
3	RX
2	TX
1	+

AF-H7E Lite Pinout

• Pinout (2/4)

E GPS 1

SERIAL3 · 6-Pin · JST-GH

6	—
5	SDA
4	SCL
3	RX
2	TX
1	+

F DEBUG

SERIAL7 · 6-Pin · JST-SH

6	—
5	SWCLK
4	SWDIO
3	RX
2	TX
1	VREF

G TELEM 2

SERIAL2 · 6-Pin · JST-GH

1	+
2	TX
3	RX
4	CTS
5	RTS
6	—

H GPS 2

SERIAL5 · 6-Pin · JST-GH

1	+
2	TX
3	RX
4	SCL
5	SDA
6	—

I USB-C

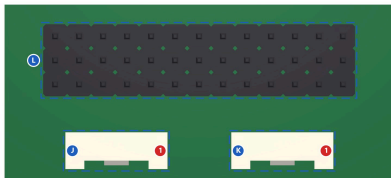
Type-C

USB 2.0

AF-H7E Lite Pinout

• Pinout (3/4)

Rear half — top side: POWER and PWM (rear edge at the bottom)



J POWER 1

6-Pin · Micro-Lock Plus · pin 1 = right end (Molex SD-505568-001)

6	5	4	3	2	1
—	—	SDA	SCL	+	+

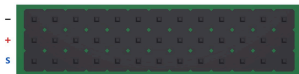
K POWER 2

6-Pin · Micro-Lock Plus · pin 1 = right end (Molex SD-505568-001)

6	5	4	3	2	1
—	—	SDA	SCL	+	+

L PWM OUT — 3 × 13 header · SB = SBUS OUT

SB = left, M1 = right · rows as seen from above: —, +, S (front → rear) · + = servo rail · 2.54 mm pitch

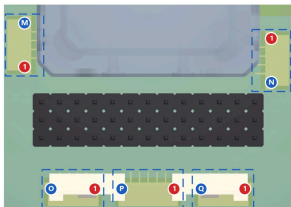


—	—	—	—	—	—	—	—	—	—	—	—	—	—
+	+	+	+	+	+	+	+	+	+	+	+	+	+
S	SB	M12	M11	M10	M9	M8	M7	M6	M5	M4	M3	M2	M1

AF-H7E Lite Pinout

• Pinout (4/4)

Rear half — bottom-side connectors, seen through the PCB (rear edge at the bottom; the cream b



M I2C A

4-Pin · JST-GH

4	—
3	SDA
2	SCL
1	+

N I2C B

4-Pin · JST-GH

1	+
2	SCL
3	SDA
4	—

O SERIAL4

SERIAL4 · 4-Pin · JST-GH · UART8 - default GPS

4	3	2	1
—	RX	TX	+

P RC IN

5-Pin · JST-GH · RC input · SBUS / PPM / DSM (auto-detected)

5	4	3	2	1
—	3V3	RSSI	RC_IN	+

Q SERIAL6

SERIAL6 · 4-Pin · JST-GH · UART4

4	3	2	1
—	RX	TX	+

AF-H7E Lite Pinout

• Pinout Notes

Preliminary pin definition — AF-H7E Lite is in development and connectors may change before release. On every JST connector pin 1 is the supply pin and the last pin is GND, following the Pixhawk connector standard.

Each serial port is engraved with its ArduPilot number under the port name, such as TELEM 1 with SERIAL1 | UART7 below it. Only the ports we recommend for a purpose carry a name — the ones with an electronic fuse (TELEM 1 · 2, GPS 1 · 2, RC IN). The general-purpose rear ports are simply marked SERIAL4 and SERIAL6 with their MCU peripheral underneath.

Connector names, serial numbers and MCU peripherals are three different things: the rear 4-pin port marked SERIAL4 runs on MCU UART8, and SERIAL6 on MCU UART4. The Serial Mapping row in the specifications lists all of them, taken from SERIAL_ORDER in the board definition. TELEM 1 and TELEM 2 default to MAVLink telemetry.

AF-H7E Lite Pinout

GPS 1 and GPS 2 are 6-pin Pixhawk GPS ports (UART + I2C on one cable) placed on the left and right edges at the same height, so a redundant pair sits symmetrically on the airframe; each also works as a plain serial port. I2C A and I2C B are separate 4-pin ports for devices that only speak I2C (airspeed, rangefinder, external compass, LED); I2C A shares its bus with GPS 1 and I2C B with GPS 2, and further devices can be daisy-chained on a splitter cable as long as their addresses differ. There is no GPS/safety port and no safety switch.

RC IN uses the AF-H7E 5-pin pinout and takes an SBUS, PPM or DSM receiver on pin 2 (protocol auto-detected, wired straight to the flight-controller MCU). Pin 1 supplies 5 V. Pin 4 is a switched 3.3 V rail for DSM satellites; the flight controller power-cycles it to bind.

On-board without a connector: microSD card slot (logging), buzzer and RGB status LED.

AF-H7E Lite Pinout

The + rail of the PWM header is not powered by the flight controller. The 13th header column, SB, is an SBUS output, not a PWM channel: it carries servo channels 1–16 on one wire from USART6 (SERIAL8) with the signal inversion done inside the STM32H7, so SBUS servos, SBUS-to-PWM decoders and gimbals plug in with a standard servo lead and take power from the servo rail. Outputs share rate and protocol within the timer groups M1–M4, M5 · M6 · M9 · M10, M7–M8 and M11–M12; DShot works on every group except M7–M8, whose timer has no DMA (PWM and OneShot only).

AF-H7E Lite Pinout

POWER 1 and POWER 2 sit on the top side at the rear edge with their latch facing the rear wall; the rear wall is no higher than the roof, so the latch is pressed from above with nothing in the way; RC IN, SERIAL4 and SERIAL6 sit under them on the bottom side and plug in from the rear like the side connectors. The PWM header sits 2.3 mm behind the sensor module, with a 1 mm aluminium lip in between: as on AF-H7E the signal row (S) is the rearmost row and the case has a keyed comb behind it (the key slots are closed at the back), so servo plugs only fit with S at the rear. The pinout image is a top view of the concept carrier: side connectors are mounted on the bottom side and plug in from the edges, and each table lists pins in the order they sit when seen from above. POWER 1 and POWER 2 use the same Molex Micro-Lock Plus connector as AF-H7E; their pin-1 end is still to be confirmed. POWER 1 and POWER 2 take I2C (INA2xx) power modules; a DroneCAN power module (for example 14S / 200 A) reports over CAN 1 or CAN 2 (BATT_MONITOR 8) and needs a split cable that feeds its 5 V into POWER 1, because the CAN connectors supply 5 V rather than accept it.

AF-H7E Lite Pin Definition

• POWER 1

Molex Micro-Lock Plus 6P (1.25 mm) · same as AF-H7E ·
Primary power input

Pin	Signal	Function
1	VCC_IN	5 V supply input from the power module
2	VCC_IN	5 V supply input from the power module
3	SCL	Power-module I2C clock (voltage / current monitor)
4	SDA	Power-module I2C data (voltage / current monitor)
5	GND	Ground
6	GND	Ground

AF-H7E Lite Pin Definition

• POWER 2

Molex Micro-Lock Plus 6P (1.25 mm) · same as AF-H7E ·
Redundant power input

Pin	Signal	Function
1	VCC_IN	5 V supply input from the second power module
2	VCC_IN	5 V supply input from the second power module
3	SCL	Power-module I2C clock (voltage / current monitor)
4	SDA	Power-module I2C data (voltage / current monitor)
5	GND	Ground
6	GND	Ground

AF-H7E Lite Pin Definition

• TELEM 1

JST-GH 6P · SERIAL1 · UART7 · default MAVLink2

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	CTS	Clear to send — hardware flow control input
5	RTS	Request to send — hardware flow control output
6	GND	Ground

AF-H7E Lite Pin Definition

• TELEM 2

JST-GH 6P · SERIAL2 · UART5 · default MAVLink2

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	CTS	Clear to send — hardware flow control input
5	RTS	Request to send — hardware flow control output
6	GND	Ground

AF-H7E Lite Pin Definition

• GPS 1

JST-GH 6P · SERIAL3 · USART1 + I2C2 · default GPS

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	SCL	I2C clock — compass in the GPS module (bus shared with I2C A)
5	SDA	I2C data
6	GND	Ground

• SERIAL4

JST-GH 4P · rear edge · SERIAL4 · UART8 · default GPS

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	GND	Ground

AF-H7E Lite Pin Definition

• GPS 2

JST-GH 6P · SERIAL5 · USART2 + I2C3 · default GPS

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	SCL	I2C clock — compass in the GPS module (bus shared with I2C B)
5	SDA	I2C data
6	GND	Ground

• SERIAL6

JST-GH 4P · rear edge · SERIAL6 · UART4

Pin	Signal	Function
1	VCC	5 V output
2	TX	UART transmit (FC → device)
3	RX	UART receive (device → FC)
4	GND	Ground

AF-H7E Lite Pin Definition

• I2C A

JST-GH 4P · I2C2

Pin	Signal	Function
1	VCC	5 V output
2	SCL	I2C clock — external compass, rangefinder, airspeed ...
3	SDA	I2C data
4	GND	Ground

• I2C B

JST-GH 4P · I2C3

Pin	Signal	Function
1	VCC	5 V output
2	SCL	I2C clock — external compass, rangefinder, airspeed ...
3	SDA	I2C data
4	GND	Ground

AF-H7E Lite Pin Definition

• CAN 1

JST-GH 4P · CAN1 · DroneCAN

Pin	Signal	Function
1	VCC	5 V output to CAN peripherals (not a power input)
2	CAN_H	CAN bus high
3	CAN_L	CAN bus low
4	GND	Ground

• CAN 2

JST-GH 4P · CAN2 · DroneCAN

Pin	Signal	Function
1	VCC	5 V output to CAN peripherals (not a power input)
2	CAN_H	CAN bus high
3	CAN_L	CAN bus low
4	GND	Ground

AF-H7E Lite Pin Definition

• RC IN

JST-GH 5P · rear edge · RC input · SBUS / PPM / DSM (auto-detected)

Pin	Signal	Function
1	VCC	5 V receiver supply
2	RC_IN	SBUS / PPM / DSM receiver input (protocol auto-detected)
3	RSSI	Analog RSSI input (0–3.3 V)
4	VCC_3V3	Switched 3.3 V supply for DSM / Spektrum satellite receivers (power-cycled to bind)
5	GND	Ground

• ETHERNET

JST-GH 4P · 100BASE-T

Pin	Signal	Function
1	RX-	Ethernet receive pair – (same pin order as AF-H7E)
2	RX+	Ethernet receive pair +
3	TX-	Ethernet transmit pair –
4	TX+	Ethernet transmit pair +

AF-H7E Lite Pin Definition

• DEBUG

JST-SH 6P · Pixhawk Debug Mini · SERIAL7 · USART3 · console + SWD

Pin	Signal	Function
1	VREF	3.3 V reference for the debug probe
2	CONSOLE_TX	Debug console transmit (FC → probe)
3	CONSOLE_RX	Debug console receive (probe → FC)
4	SWDIO	SWD data — bootloader programming and firmware recovery
5	SWCLK	SWD clock
6	GND	Ground

AF-H7E Lite Pin Definition

• USB

USB Type-C · SERIAL0 · USB 2.0 Full Speed

Pin	Signal	Function
A4 B4 A9 B9	VBUS	5 V from USB — configuration and firmware update on the bench
A6 B6	D+	USB data +
A7 B7	D-	USB data -
A5 B5	CC1 / CC2	Configuration channel (device role)
A1 B1 A12 B12	GND	Ground

• PWM OUT M1-M12

2.54 mm 3 × 13 pin header · columns 1-12 · Main outputs 1-12

Pin	Signal	Function
S	M1 ... M12	Motor / servo output signal, one per channel (PWM · OneShot · DShot; M7-M8 without DShot)
+	V_SERVO	Servo rail, bussed across all channels — supplied externally (e.g. BEC)
-	GND	Ground

AF-H7E Lite Pin Definition

• SB (SBUS OUT)

2.54 mm 3 × 13 pin header · column 13, next to M12 · SERIAL8
· USART6 · SBus servo out

Pin	Signal	Function
S	SBUS_OUT	SBUS output — servo channels 1–16 on one wire (SERIAL8_PROTOCOL 15, SERIAL8_OPTIONS 2 inverts TX inside the MCU)
+	V_SERVO	Servo rail, bussed across all channels — supplied externally (e.g. BEC)
–	GND	Ground

AF-H7E Lite Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`. Mounting: 16 × 30 mm / M3 (4 blind holes in the base, 3 mm thread).
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 4.75 – 5.7 V (Rated 5 V).
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-H7E Lite Firmware

• Available Firmware

Type	Version	Date
ArduPilot Copter (.apj package)	0.1.0	2026-09-17
ArduPilot Copter - Bootloader + App (merged HEX / DFU / SWD)	0.1.0	2026-09-17
ArduPilot Plane (.apj package)	0.1.0	2026-09-17
ArduPilot Plane - Bootloader + App (merged HEX / DFU / SWD)	0.1.0	2026-09-17

Download files from the product page (QR).

AF-H7E Lite Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Product in development — the pin definition may change before release; check the latest catalog.
- Supply the PWM servo rail (+) from an external BEC.
- Firmware v0.1.0 is a preliminary, build-verified release. Copter and Plane share board ID 6207 — pick the image by file name.

AF-H7E Lite Safety Notes



<Detailed Manual>

